Department of Economics

MSc Extended Essay Cover Sheet

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COURSE CODE	E	C	4	2	3

TITLE	**Industrial robots and crime: Assessing the impact of automation on local US crime rates.**

WORD COUNT	5	6	6	6

Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered. Binding is not necessary, however it is crucial that all pages are secure, i.e. stapled.*

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Industrial robots and crime: Assessing the impact of automation on local US crime rates.

Abstract

As technological advancement continues to shape labour markets across the globe, it is imperative to understand the broader consequences of replacing tasks previously performed by humans, with industrial robots. This paper employs the shift share instrumental variable framework pioneered by Acemoglu and Restrepo (2020) to explore the relationship between automation and crime rates in the United States (US), motivated by concerns over endogeneity in US robotics adoption. The forthcoming analysis reveals some interesting insights: whilst no discernible impact on the immediate change in crime rates is found, subsequent years (2000-2014 and 2001-2015) suggest a notable increase in both property and violent crime rates. This delay points towards the intricate dynamics of automation-induced job displacement and its repercussions on criminal activity. However, due to data and methodological constraints, this should be regarded as suggestive evidence, motivating future research into dynamic effects of robots.

1. Introduction

The relationship between labour market conditions, particularly unemployment, and crime is a well-established field of economic research which grew following Becker's seminal analysis in 1968 (Becker, 1968). There has been a particular focus on the effects of unemployment on property crime (Sampson and Laub, 1997; Gould et al., 2002), where property crimes are viewed as being economically motivated and are therefore responsive to local economic conditions. The advent and rapid adoption of industrial robots presents a novel challenge, on one hand offering potential productivity benefits and stimulating economic growth, but also threatening to displace workers. The growing literature on the labour market effects of automation suggest two potential mechanisms via which industrial robot adoption may influence crime rates: firstly, by altering aggregate labour market conditions through increased unemployment and reduced wages, and secondly, by exacerbating economic inequality due to differential impacts along the income distribution, across occupations and across industries. Policymakers must carefully consider potentially adverse impacts of automation induced by technological progress, which may extend beyond short-term displacement of workers, also leading to increases in criminal activity due to both job losses and regressive wage effects.

This paper studies a panel dataset of 500+ commuting zones (CZs) and follows Acemoglu and Restrepo's (2020) (henceforth, AR (2020)) shift-share instrumental variable (SSIV) strategy, exploiting plausibly exogenous industry-level technological shocks, to investigate the effects of automation on crime rates in the United States (US). Shocks to US industry robotics adoption are instrumented using robotics adoption in five European countries, which are weighted by local US occupation structure to generate cross-sectional variation. Local labour markets are defined by 1990 CZs (Tolbert and Sizer, 1996)[1], following the recent literature investigating labour market effects on crime within CZs (Bell et al, 2022; Bell et al, 2016). This strategy aims to reduce the likelihood of within-unit heterogeneity confounding the link between robotics adoption and crime, as seen in previous state-level studies such as Raphael and Winter-Ebmer (2001). It is expected that in the short term, displacement of workers and skill polarisation that can arise from automation will contribute to an increase in crime rates, with a more pronounced effect on property crimes due to the presence of economic incentives. This hypothesis is consistent with US-based evidence that robotics adoption has negative aggregate employment effects (AR, 2020).

[1] Commuting zones (CZs) are weakly related by construction. They combine neighbouring county pairs with strong commuting ties to capture the spatial proximity of these regions.

In the longer run, as productivity benefits materialize and new jobs are created elsewhere, this effect may decline (Moretti, 2023).

2. Existing literature

The relevant literature for this paper can be divided into two strands and will be accordingly discussed: (1) the effect of automation on labour market outcomes and (2) the effect of labour market conditions on crime.

2.1. Effect of robots on labour market outcomes

The existing research on robots' impact on labor market outcomes is mixed. Graetz and Michaels' (2018) pioneering work exploits within-sector variation in industrial robot adoption across 17 developed countries to investigate the impact on employment and wages over 1993-2007. They use an SSIV strategy similar to AR's (2020) subsequent instrument for measuring US robotics exposure, weighting sectoral robot adoption by within-country employment shares. Their findings suggest skill polarising effects of automation, where low skilled workers' employment share declines relative to that of middle and high-skill workers, but they find no adverse effects of automation on aggregate wages or employment. In contrast, AR (2020) find that robots have negative aggregate employment and wage effects in the US, in addition to a disproportionate effect on low-skill workers.

AR (2020) introduced a novel instrument for robotics exposure, leveraging shocks to robotics in European countries which are more advanced than the in automation. This method has been adopted in several recent papers. Most recently, Albinowski and Lewandowski (2023) focus on the varying effects of robotics adoption across demographic groups in the labour market. They find that while robots increase the share of young and 'prime age' women in employment, they decrease employment shares of 'prime age' males and older women. Regarding aggregate labour market outcomes, the focus of this paper, Dauth et al (2021) discover that, in Germany, worker displacement in manufacturing was offset by job creation in services. In contrast, Bachmann et al (2022) find that in European countries with low labour costs, robot adoption in fact reduced unemployment. Effects of robotics adoption, therefore, appear highly dependent on region, suggesting that the results from this current analysis cannot be easily extrapolated beyond the US context.

2.2. Effect of labour market conditions on crime

Becker's (1968) theory proposes that an agent chooses to engage in criminal activity if there is a net expected benefit from committing the crime, after considering both illegal and legal market returns, as well as the (probabilistic) penalty if caught. If local labour demand falls and legal market returns subsequently decrease, the opportunity cost of crime falls, and one may expect crime rates to increase. As noted by Levitt (2004), violent crime is rarely economically motivated. Therefore, this theory provides a clearer rationale for why unemployment may contribute to property crime. Consistent with this, Raphael and Winter-Ebmer (2001) attribute a substantial proportion of the 1990s decline in property crime rates to the decline in unemployment but find weak evidence for violent crime at the US state-level. Similarly, Lin (2008) fails to find any significant effect on violent crime.

However, a positive short-term effect of automation on violent crime cannot be disregarded. Blau and Blau (1982) find that in the US' largest metropolitan areas, areas with greater economic inequality, in particular greater interracial socioeconomic inequality, were associated with higher violent crime rates in 1970. Their analysis relied on ordinary least squares (OLS) regression, which makes causal inference challenging. However, in more recent work, Fajnzylber et al (2002) investigated this link causally, instrumenting the explanatory variables using appropriate lags to address two-way links between unemployment and violent crime. Their findings indicate significant positive effects of income inequality, measured by the Gini index, on homicide and robbery rates, which are robust to sample selection, measures of inequality and estimation methods.

The likelihood of omitted variable bias is high due to macroeconomic conditions that simultaneously influence both unemployment and crime, potentially leading to reverse causality. Consequently, many studies have employed instrumental variable (IV) strategies, such as Sampson, Raudenbush, and Earls (1997), or utilised exogenous shocks to local labour markets to disentangle the effects of unemployment on crime. Trade shocks, for example, have been exploited in various studies (Dix-Carneiro et al., 2018; Allen & Sawyer, 2024). Similarly, to address endogeneity concerns, this paper leverages AR's (2020) modelling of a technological shock to CZs, which they find to have negative employment effects, to investigate the impact of automation on local property and violent crime rates.

3. Data

3.1. Crime data and sample selection

The crime data used in this paper is sourced from the Offences Known and Cleared dataset of the Uniform Crime Reporting (UCR) Program, and has been cleaned by Jacob Kaplan (Kaplan, 2021). The dataset reports agency-level counts for offences which actually occurred (i.e. they were not unfounded crimes). Following the literature, this paper examines two key measures of interest: property crime and violent crime. Property crime is an aggregate measure of burglary, arson and theft, and violent crime aggregates counts for murder, manslaughter, rape, aggravated assault and robbery. Crime counts are aggregated across agencies at the county level, converted into rates and subsequently mapped to CZs.

A significant challenge of the data is the high level of non-reporting by agencies, which persists throughout this paper's period of interest, from 1993-2017. Following Bell et al (2022), agencies which provide a full set of 12 monthly observations are selected and used to calculate a county-level population coverage ratio. Counties are selected into the sample based on whether they exceed a coverage threshold of 95%. As a robustness check, an alternative sample is selected using a coverage indicator initially developed by the Inter-university Consortium for Political and Social Research (ICPSR) and used in Fone et al's (2019) study of minimum wage effects on crime. The results are robust to this selection.

Since this paper is interested in the long-run effect of an automation trend on crime rates, log-transformed rates are differenced over the 1993-2007 period[2]. Due to the long differencing of crime data in this analysis, the selection of counties described above results in a substantial sample size reduction, since a missing report in 1993 or 2007 (or both) generates a missing observation. The sample, therefore, reduces from 722 CZs to 516 and 526 for violent crime and property crime, respectively. Whilst spatial and temporal proxying approaches and interpolation methods were considered to impute missing observations, these approaches are generally unsuitable due to high geographic and temporal heterogeneity of unobservable determinants of crime.

The sample selection of CZs could be biased if the CZs excluded due to non-reporting at the county-level systematically differ from those included. For example, it is possible that more urbanised regions, which may

[2] 1993-2007 for the main analysis, and 10 additional windows in section 5.3.

experience similar crime rate shifts over this period, have both higher levels of technological adoption and better reporting practices. Therefore, these regions are more likely to be included in the sample than rural CZs, biasing the results. This concern is investigated by comparing summary statistics (see Appendix table A.1) of key variables across CZs excluded from the sample to those included. Whilst the means of these variables appear to differ substantially across the two groups, simple t-tests confirm that there are no statistically significant differences across the two groups of reporting vs non-reporting CZs.[3]

3.2. Robotics exposure

The measures for robotics exposure used in this analysis are taken directly from AR (2020) and use US predicted adoption data for the OLS specification and foreign robotics penetration of five European countries (Denmark, Finland, France, Italy and Sweden) for the instrumental variables two-stage least squares (IV 2SLS) specification. The strategy is described in more detailed in section 4.

3.3. Control variables

Covariates selected for the analysis are based on the mechanisms via which automation may affect crime rates. Household median income is sourced from 1990 US Census data and unemployment rates for 1990 are taken from the Bureau of Labour Statistics (BLS). Since there are many missing observations for 1990 county-level unemployment rates, unemployment data from 2000 (taken from the United States Department for Agriculture's Economic Research Service (USDA ERS)) are used to compute an average change from 1990-2000 and impute the missing 1990 values. Although an expected mechanism via which automation can affect crime rates is worker displacement, it is important to control for persistent differences in unemployment across regions which may influence crime rate trends. CZ population density (thousands of persons per square mile) is computed using county land area data also from the 1990 census. Population counts which are used as sample weights, and the remaining covariates – share of females, ethnicity and education shares and Chinese import exposure – are taken from AR (2020)'s replication package. Education shares include percentages of those who have completed high school, have received some college education, completed a college degree and completed a master's degree. Since the period of analysis spans China's accession to the World Trade Organisation, it is important to also control for potential effects of the rise in import competition which may confound the link between robotics

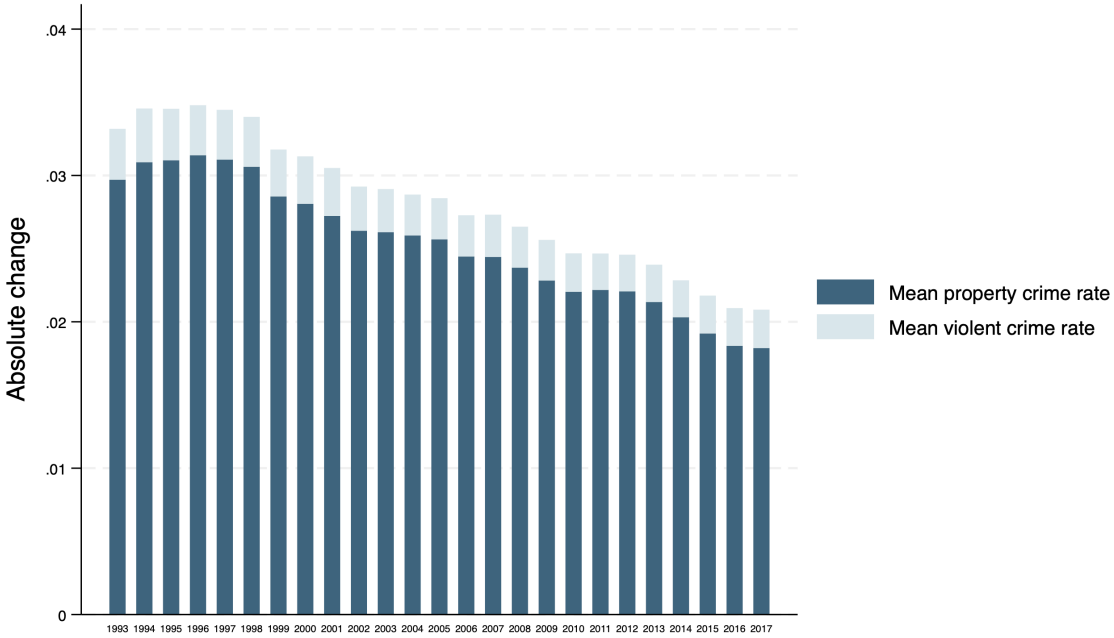
[3] At the conventional significance level of 5%.

adoption and crime rates.

3.4. Descriptive statistics

Figure 1 shows that, following a moderate increase in total crime rates from 1993-1996, there has been a sustained decline since 1996. This has been driven by a reduction in property crime rates, which accounts for the majority of offences. Table 1 below details summary statistics for the CZ sample, focusing on the 1993-2007 long-differenced crime rates and 1990 baseline covariates.

Figure 1: Change in crime rates (1993-2017)



Crime rates are measured as number of crimes committed per 1000 people.

Table 1: Descriptive statistics

Variable	Mean	Std. Dev.	Min	Max
(Log) change in property crime rates				
1993-2007	-.161	.547	-2.408	1.910
1994-2008	-.223	.529	-2.66	1.577
1995-2009	-.283	.526	-2.549	2.403
(Log) change in violent crime rates				
1993-2007	-.065	.636	-2.932	2.394
1994-2008	-.132	.615	-1.965	3.029
1995-2009	-.103	.651	-2.27	3.942
US robotics adoption	1.485	1.41	.159	11.941
Population density (persons per sq. mile)	77.292	162.528	.336	2651.599
Median income ($'000s)	22.781	4.931	11.533	45.294
Unemployment (%)	5.457	2.911	.986	28.975
High school	.619	.081	.349	.821
Some college	.248	.048	.118	.458
College	.101	.031	.039	.233
Masters	.032	.012	.014	.106
Hispanic	.057	.118	.001	.939
White	.895	.117	.38	.996
Black	.076	.116	0	.615
Asian	.008	.012	0	.157

Figure 2: Geographic variation in US robotics exposure (1993-2007)

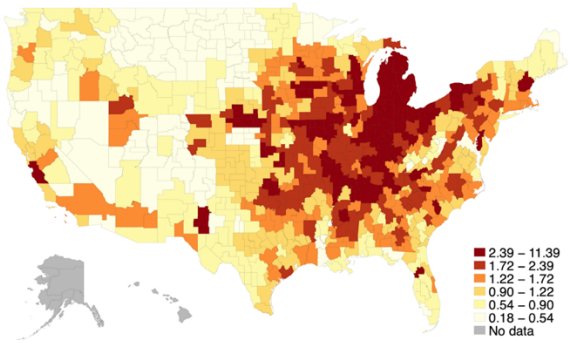


Figure 3.1: Δ Violent crime rate

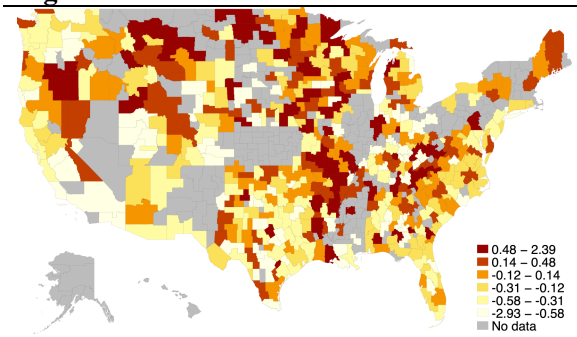
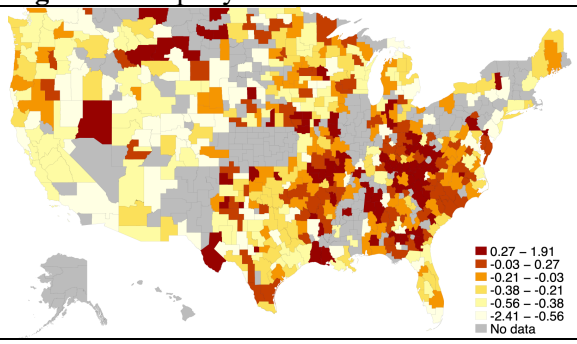
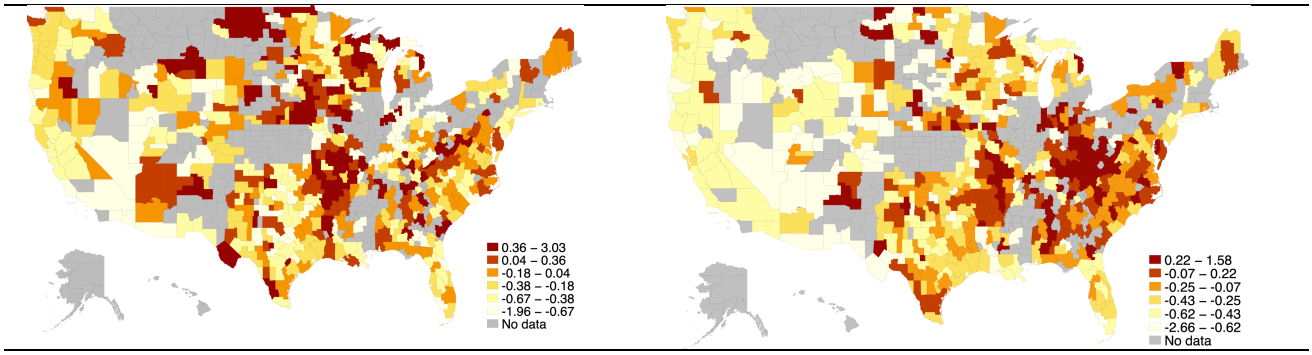


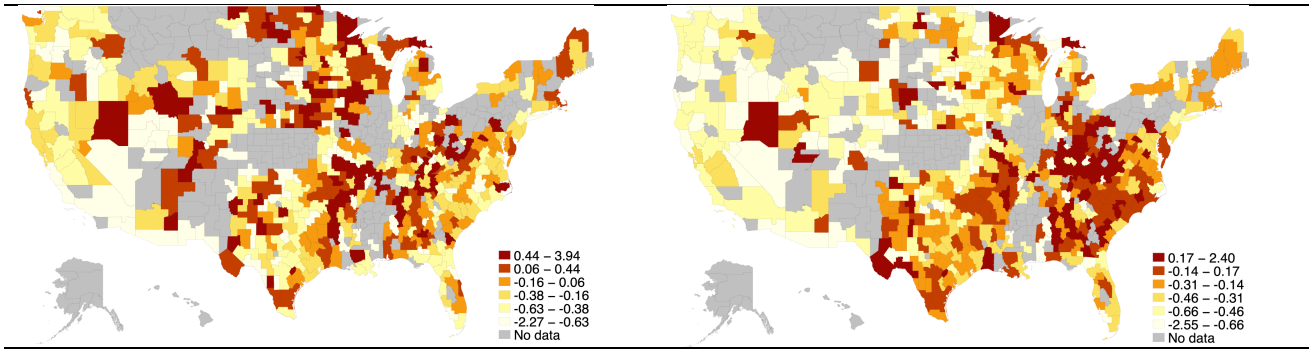
Fig 3.2: Δ Property crime rate



(1993-2007)



1994-2008



1995-2009

3.5. Geographic evidence

Figures 3.1 and 3.2 illustrate the geographic variation across CZs in the change in property and violent crime rates, respectively, over 1993-2007, 1994-2008 and 1995-2010. Figure 1.1, replicated from AR (2020), shows the variation in robotics exposure and is included for comparison. The regions where increases in property crime are concentrated are predominantly located in the east, similar to regions of higher robotics exposure, whereas changes in violent crime appear to be more randomly distributed. CZs with missing data do not appear to change much. Although the spatial clustering of non-reporting CZs raises some concern, since groups of neighbouring zones are likely to have similar economic and demographic characteristics and could therefore differ systematically from included CZs, this concern has been partially alleviated by the comparison of observable characteristics described in section 3.1. Systematic differences in unobservables cannot be tested, which emphasises the importance of sample selection robustness checks.

4. Empirical Strategy and Theory

4.1. Main specification

The main specification is as follows:

$$\Delta \ln Y_c = \beta\, Robots_c + \gamma_k X_c + \alpha_r + \varepsilon_c$$

This paper is primarily interested in the long-run impact of automation on crime rates. Therefore, the outcome

variables of interest, denoted $\Delta \ln Y_c$, are the log change in property and violent crime rates in a CZ c over 14 years. The main period of interest is 1993-2007, but later periods are also investigated.

$Robots_c$ represents the long-run trend in robotics exposure defined and computed by AR (2020). This uses US (predicted) adoption data for the OLS specification and foreign robotics penetration for the IV 2SLS specification. The measure of US robotics exposure is defined as $Robots_c = \Sigma_{i \epsilon I} s_{ci}^{1990} \cdot g_i$, where g_i represents the industry-level shock to robotics penetration in the US, accounting for the growth rate of industry output. $\Sigma_{i \epsilon I} s_{ci}^{1990}$ represents baseline employment shares in 1990. In the IV specification, $Robots_c$ is replaced with the instrument $z_c = \Sigma_{i \epsilon I} s_{ci}^{1970} \cdot \tilde{g}_i$, where $\tilde{g}_i$ represents a foreign robotics penetration measure, computed using data from five European countries which are considered to be ahead of the US in automation technology. The employment shares s_{ci} are replaced with 1970 data, in order to focus on historical differences in local industry structure. This aims to make the instrument exogenous with respect to regression covariates measured in 1990. Importantly, however, this "delocalization" of shares does not make the instrument causally prior with respect to unobservable determinants of $\Delta \ln Y_c$ (Ferri, 2022), an issue discussed in section 4.3.1. X_c denotes the vector of covariates described in section 3.3, as well as CZ 1990 major industry shares. These are included to capture persistent differences in local economic structure, which may contribute to criminal activity independently of technological change. α_r denotes census division fixed effects and ε_c represents idiosyncratic, unobservable CZ characteristics which partially determine the long-run shift in crime rates. Including covariates as baseline measures increases the precision of estimates. More importantly, using predetermined levels of unemployment and education levels reduces the likelihood of reverse causality, in which areas characterised by high crime increase may experience a reduction in employment opportunities, increasing unemployment and reducing the gains from education.

The dependent variable is lagged 10 times and tested in separate regressions (1994-2008, 1995-2009, …, 2003-2017), a strategy which is adopted for two reasons. Firstly, US robotics exposure data at the industry level are only available from 2004, and as a result, AR's (2020) US shock measure (which has been scaled to a 14-year horizon for the analysis) may only capture adoption patterns from 2004-2007. If the shock is only measured from 2004, a contemporaneous impact of the 1993-2007 automation trend may not materialise in the regression of 1993-2007 crime rates. Alternatively, if timing of shock measurement is not a concern, examining later crime rate trends could reveal delayed effects of automation on crime rates. The consequences of job displacement and

increased inequality might not be immediately reflected in crime rates.

4.2. Econometric issues

The likelihood of omitted variable bias is high in this analysis. Three key concerns are identified which threaten the validity of OLS estimates. Firstly, reverse causality may occur, where businesses in CZs with high crime rates are deterred from investing in robotics technology due to fears of theft or damage. For instance, Ashby and Ramos (2013) found that foreign direct investment decreases in response to organised crime in Mexico. Secondly, a local economic shock, such as a recession, could simultaneously influence crime rates (positively) and robotics adoption (negatively). Lastly, industries may adopt robotics as a result of policy shifts that also impact crime rates. For example, a relaxation of trade barriers could trigger competitive practices and affect crime through changes in local economic conditions, such as employment opportunities. This motivates the SSIV strategy employed by AR (2020), in which foreign robotics penetration is used to instrument for US adoption. The necessary assumptions and plausibility are discussed in detail below.

4.3. SSIV identification

4.3.1. Identification from shares versus shocks

The SSIV literature is characterised by two identification strategies; share exogeneity and shock exogeneity. Borusyak et al (2022) (henceforth, BHJ (2022)) proposes that the appropriate strategy should be decided ex ante, based on economic context.

Goldsmith-Pinkham et al (2020)'s share exogeneity approach requires that the pre-determined industry composition of CZs, as measured by 1970 employment shares, cannot predict unobserved crime shocks in 1993 and 2007. Formally, this condition can be written as:

$$E[\varepsilon_c | s_{ci}] = 0$$

This assumption lacks credibility, particularly in regions where the occupation structure has persisted since 1970 (for example, Texas as the US' energy capital, Michigan as an automotive hub, New York as a global financial centre). It is plausible that some local economic shock in residual ε_c, which drives crime rates, results from regional industry composition s_{ci}. For example, a downturn specific to the automotive industry could lead to economic hardship in a CZ which has persistently specialised in this sector and increase local crime rates. Whilst

this leads to a violation of the exogeneity condition above and renders the share identification approach ineffective, fortunately, BHJ's (2022) orthogonality conditions derived from their shock-level equivalence result are more tenable, and can be used to infer causal estimates.

4.3.2. Shock-level equivalence

BHJ's (2022) shock-level equivalence result is first briefly described, from which a necessary orthogonality condition for validity and law of large numbers (LLN) for consistency follow. Firstly, it is assumed that ε_c is orthogonal to vector X_c in (1): unobserved determinants of the change in crime rates are uncorrelated with predetermined, historic differences in demographic and socioeconomic characteristics across CZs. Sample residuals $\Delta \ln Y_c{}^{\perp}$ and $Robots_c{}^{\perp}$ can be obtained from regressing $\Delta \ln Y_c$ and $Robots_c$ separately on X_c; the control variables are partialled out. Then, by transforming the residualised variables as shock exposure-weighted averages, the main regression equation (1) becomes:

$$\overline{\Delta \ln Y_i}^{\perp} = \alpha + \beta \overline{Robots_i}^{\perp} + \bar{\varepsilon}_i^{\perp}$$

This transformation effectively translates crime rates and other covariates to the industry level (the identifying level of variation), where they would not conventionally be observed. BHJ (2022) offers the following interpretations: $\overline{\Delta \ln Y_i}^{\perp}$ is the average residualised outcome of the CZ most exposed to a robotics shock in industry i, and $\bar{\varepsilon}_i^{\perp}$ represents the average unobserved determinants of crime in CZs where industry i employs the most workers. Instrumenting $\overline{Robots_i}^{\perp}$ using $\tilde{g}_i$, European robotics penetration, now gives the SSIV estimator β. As emphasised by BHJ (2022), this result motivates their identification approach but does not alter the interpretation of β, offering the warning that although the SSIV estimator can be computed at industry levels, it may still capture spillovers across CZs.

4.3.3. Identifying assumptions

1. **A1. Weak shock exogeneity:**

It is assumed that instrumented shocks to robotics penetration across all 19 industries have the same expected value when conditioning on industry-level unobservable determinants of crime rate changes, $\bar{\varepsilon}$, and average employment share, s (which captures the average importance of an industry shock across the entire sample). Formally, $E[\tilde{g}_i|\bar{\varepsilon}, s] = 0$ for all industries i, which guarantees unbiasedness of $\hat{\beta}$ given that the IFR exposure

shares sum to 1 in every CZ[4]. European robotics adoption must not be chosen strategically based on trends in US adoption, which is correlated with unobserved determinants of crime rate shifts as explained in 3.2. This is plausible, since by choosing countries that are more advanced than the US, it is unlikely that imitation or competitive cost-cutting measures are introduced in Europe as a result of changes in industry practices in the US. This weak industry-level orthogonality condition provides a condition equivalent to the exclusion restriction (BHJ, 2022).

2. **A2. Many uncorrelated shocks**

A2 requires that (i) there are sufficiently many industries such that the expected Herfindahl index of average robotics shock exposure, $E[\Sigma_i s_i^2]$, converges to zero, and (ii) that shocks are uncorrelated, conditional on exposure shares and unobservable determinants of $\overline{\Delta \ln Y_i}^{\perp}$: $Cov[g_i, g_{i'}|\bar{\varepsilon}, s] = 0$ for all pairs of industries (i,i') where $i \neq i'$. The equivalence result allows the application of a shock-level LLN and guarantees consistency of the estimator.

3. **A3. Relevance of instrument**

Having discussed instrument validity, the final critical assumption for identification from a 2SLS approach is that the instrument is sufficiently correlated with the endogenous regressor. This can be investigated by assessing strength of the first stage regression of US robotics adoption on European robotics adoption.

4.3.4. Threats to validity

A necessary condition for A1 to hold is that there cannot be some unobserved long-run factor contributing to both the increase in European robotics adoption and US crime rates. Supporting evidence for this assumption is provided in section 5.5, through a placebo test which investigates the potential correlation between robotics exposure over 1993-2007 and past trends in crime rates.

An additional threat to this analysis is that with a finite set of 19 industries, one sector dominates and prevents the asymptotic convergence in A2. For simplicity, it is taken for granted in this analysis that the 19-industry classification offered by AR (2020) provides sufficiently many uncorrelated shocks. However, future research

[4] This condition is derived from the full data moment condition in BHJ (2022).

should seek to confirm that no sector dominates by baseline shares.

5. Results

All regressions are weighted by 1990 CZ population and report standard errors (SEs) that are robust to arbitrary heteroscedasticity and spatial autocorrelation between states. The OLS-estimated coefficients in Table 2 show sensitivity to the specification of the covariates vector. Controlling for socioeconomic characteristics and education reverses the sign of the estimated effect of US robotics exposure on property crime rates, as seen in column (3) of Table 2, panel A. This reverts back upon inclusion of Chinese imports exposure. Controlling for industry trends in column (5) maintains the negative sign. The results are highly insignificant for both outcome variables of the change in property crime rates and violent crime rates.

5.1. OLS results

Table 2: OLS results (1993-2007)

Panel A: *Change in property crime rate*					
	(1)	*(2)*	*(3)*	*(4)*	*(5)*
US robotics exposure	-0.004	-0.006	0.003	-0.002	-0.008
	(0.024)	(0.017)	(0.017)	(0.016)	(0.014)
Number of observations	526	526	526	526	526
Panel B: *Change in violent crime rate*					
US robotics exposure	-0.003	0.006	0.025	0.018	0.020
	(0.016)	(0.010)	(0.014)	(0.012)	(0.016)
Number of observations	516	516	516	516	516
Controls					
Census division fixed effects	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*
Demographic controls		*Yes*	*Yes*	*Yes*	*Yes*
Socioeconomic characteristics and education			*Yes*	*Yes*	*Yes*
Chinese import exposure				*Yes*	*Yes*
Industry shares					*Yes*

** p<.01, * p<.05

Regressions specify census division regional fixed effects and are weighted by population in 1990. Columns (2)-(5) specify demographic controls (share of females, age structure, share of white/black/Asian/Hispanic), columns (3)-(5) control for socioeconomic characteristics (income and unemployment) and education, columns (4)-(5) add a control for change in Chinese import exposure and column (5) adds controls for the shares of 10 industries.

5.2. IV 2SLS results

The effect of US robotics exposure on the long-run change in crime rates is now estimated by exploiting the plausibly exogenous variation in US penetration, driven by foreign advances in automation technologies, as

described in section 3. First stage and 2SLS results are reported in Table 3 and Table 4.

Table 3: First stage results

US robotics adoption				
	(1)	*(2)*	*(3)*	*(4)*
European robotics adoption	1.047**	1.021**	1.047**	1.021**
	(0.129)	(0.128)	(0.129)	(0.128)
N	526	526	516	516

US robotics exposure is weighted by US 1990 employment shares and European robotics exposure is weighted by US 1970 employment shares.

Table 4: IV 2SLS results (1993-2007)

	Change in (log) property crime rate		Change in (log) violent crime rate	
	(1)	*(2)*	*(3)*	*(4)*
US robotics adoption	-0.017	-0.033	0.030	0.029
	(0.020)	(0.018)	(0.018)	(0.021)
Kleibergen-Paap rk Wald F statistic	66.19	57.29	66.24	57.340
N	526	526	516	516

First stage correlations reported in Table 3 show a strong correlation between the instrument and endogenous regressor. The Kleibergen-Paap F statistic, which is valid under reporting of heteroscedasticity-robust, clustered SEs, varies between 57 and 66. This confirms correlation between the excluded instrument, European robotics exposure, and endogenous variable, US robotics exposure. Columns (1) and (3) include the full set of controls in vector X, excluding industry shares, which are added in columns (2) and (4). Similar to Graetz and Michaels' (2018) empirical work, the SSIV loses some power when controlling for industry trends, but the instrument is still strong.

Focusing on the preferred specification which includes all covariates, the IV 2SLS results are consistent with OLS in terms of sign (see Table 2, column (5)). However, the coefficients describing semi-elasticities with respect to both outcome variables have increased in magnitude, especially so for property crime, in which the coefficient increased almost five-fold in magnitude, to -0.033. Since the negative coefficient for property crime and positive coefficient for violent crime both increased when instrumenting US robotics adoption, the endogeneity bias described in section 4.2 is ambiguous in direction. It is possible that the OLS estimates suffered from an

attenuation bias in addition to the econometric issues described in 3.2, due to bias induced by measurement error in US robotics exposure. The use of foreign robotics penetration aims to address this.

5.3. Dynamic effects

Results from running 10 additional regressions of 14-year changes in crime rates are displayed in Table 5 and are visualised in the coefficient plots in Figures 2 and 3. The confidence intervals are smaller in general for violent crime rates, reflecting less noisy data compared to property crime rates, resulting in more precise estimates. The substantially wider confidence interval for 1996-2010 estimated coefficient reflects a smaller sample of 471 CZs, due to a large amount of missing data in 1996 (non-reporting by agencies peaked this year). The plots reveal an interesting pattern: the estimated coefficients increase moderately over time, remaining insignificant until the 2000-2014 period, when the coefficient increases suddenly and gains significance. The coefficients then decline towards zero and are again insignificant. With the exception of the violent crime rate change over 1996-2010, the overall pattern for property and violent crime are similar, with significant, positive estimated effects of automation on an increase in crime rates over 2000-2014.

Since the outcome variable is a change which can take both positive and negative values, interpretation is not straightforward. A negative coefficient could indicate that robotics adoption reduces a growth in crime rates or increases a decline. Whilst these interpretations do not contradict each other, since both represent an improvement in local conditions, it is important to note that policy implications may differ. The estimate for 2000-2014 implies that, holding all other observable factors constant, one additional robot per thousand workers over the period 1993-2007, resulted in an approximate 6.9% increase (reduction) in the growth (decline) of property crime rates over 2000-2014.

Before inferring the presence of a delayed impact of automation on crime rates, it is important to return to the concern raised in section 4.1. It is possible that the apparent lagged effect is in fact a result of the US shock measure starting from 2004. To investigate this possibility, reduced form estimates are computed and shared in the following section.

Table 5: IV 2SLS coefficients over time

	Change in (log) property crime rates	N	Change in (log) violent crime rates	N
1993-2007	-0.033 (0.018)	526	0.029 (0.021)	516
1994-2008	-0.014 (0.066)	509	-0.036 (0.075)	500
1995-2009	-0.023 (0.017)	520	-0.021 (0.025)	511
1996-2010	-0.011 (0.053)	475	-0.258 (0.166)	471
1997-2011	0.013 (0.070)	489	0.040 (0.077)	480
1998-2012	0.009 (0.029)	492	0.037 (0.020)	486
1999-2013	-0.001 (0.018)	502	0.039 (0.029)	496
2000-2014	0.069* (0.030)	530	0.159* (0.069)	522
2001-2015	0.039* (0.017)	531	0.037* 0.026)	522
2002-2016	0.027* (0.013)	542	-0.001* (0.020)	536
2003-2017	0.018 (-0.013)	528	-0.013 (0.019)	519

Figure 4: Effect of US adoption on property crime rate change over time (IV 2SLS)

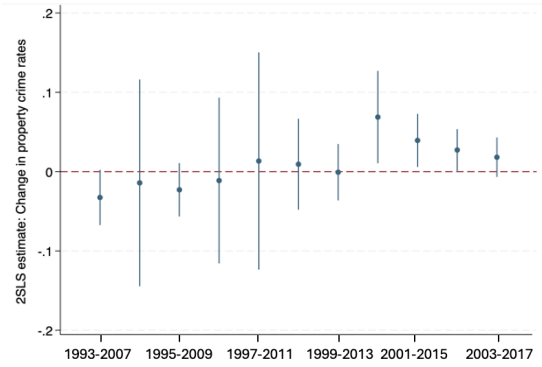
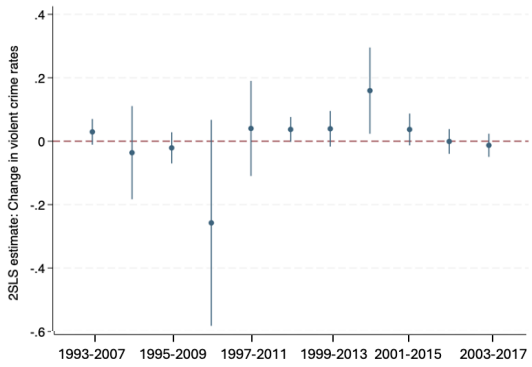


Figure 5: Effect of US adoption on violent crime rate change over time (IV 2SLS)



5.4. Reduced form results

Figure 6: Reduced form estimates of change in property crime rate

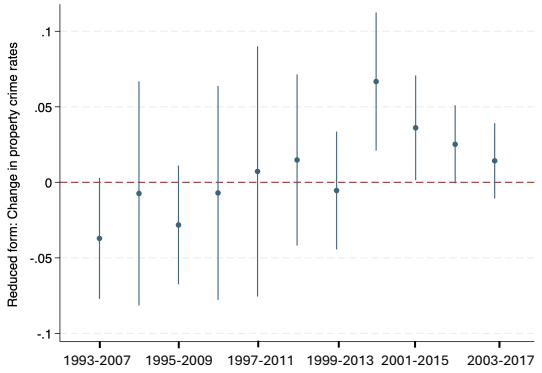
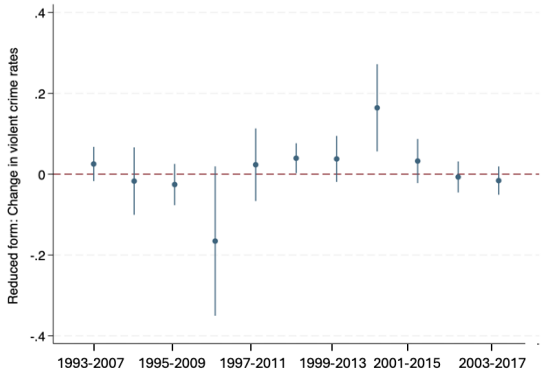


Figure 7: Reduced form estimates of change in violent crime rate



A1, or quasi-random assignment of European robotics adoption shocks, is sufficient for identification of a reduced form effect. There must not be differential trends in crime rate shifts in CZs with greater robotics exposure, such that rates would have evolved differently in the absence of robotics. Section 5.5. provides some support for this assumption. The reduced form estimates follow a very similar pattern to the 2SLS estimates, reflecting the strength of the instrument. These estimates also increase as they approach the 2000-2014 change in crime rates, which suggests that the lagged effect seen in Figures 4 and 5 is not due to delayed US shock measurement. It could instead reflect dynamic impacts of automation on shifts in crime rates occurring via mechanisms such as job displacement and worsening local economic conditions, followed by an adjustment effect as economies adapt and create new jobs in other sectors.

Whilst these results provide some interesting insights into potential longer-term effects of automation, any causal inference should be approached with caution. Since this analysis aims to exploit exogenous variation in a single measure of robotics penetration over 1993-2007, trends in crime rates beyond this horizon will not only incorporate effects of this automation trend but will be impacted by subsequent technology shocks not controlled for in the model. This makes disentangling dynamic causal effects of an automation trend even more challenging.

5.5. Placebo test

As a post-validation test to provide supporting evidence for A1, a placebo test is conducted, which uses the same 2SLS specification as the main analysis. Past changes in crime rates, which are plausibly correlated with ε_c, are

regressed on robotics exposure over the 1993-2007 period, to provide indirect evidence of instrument validity. Specifically, three pre-robotics exposure windows of crime rates changes are tested, of 1976-1990, 1997-1991 and 1978-1992. As shown in Figures 8 and 9, the estimated effects are insignificant, indicating that there is no unobserved long-run factor behind the significant coefficients found for property crime rate changes over 2000-2014 and 2001-2015, and for violent crime over 2000-2014, supporting the assumption of exogenous industry robotics shocks.

Figure 8: Placebo results – property crime

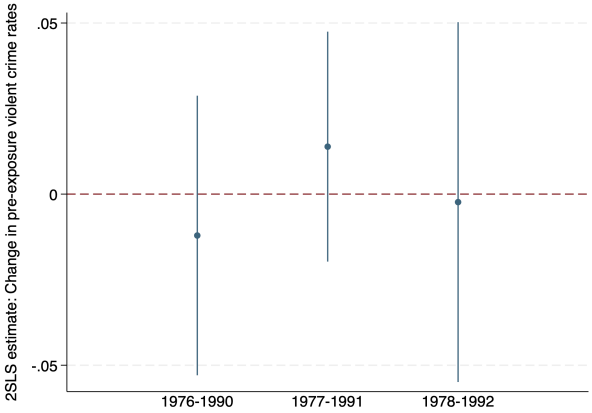
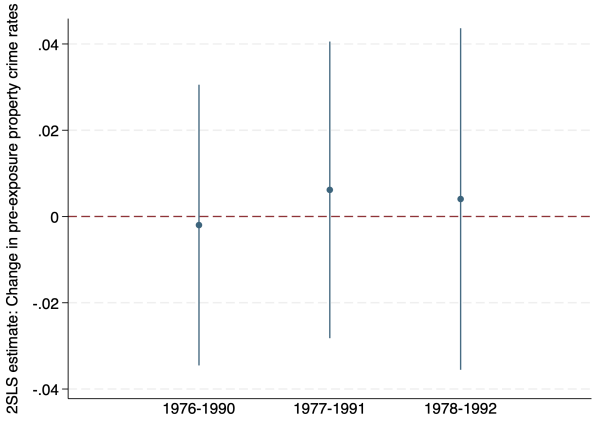


Figure 9: Placebo results – violent crime



6. Robustness

As a robustness check, a larger sample is considered in which the selection of agencies by their degree of reporting is less restrictive. A coverage ratio from the ICPR is used[5] and the 95% threshold is maintained. The results, posted in the appendix (see Tables A.2 and A.3), remain robust to this alternative sample selection, with the results for violent crime being especially similar.

7. Discussion

In order to address the challenge of endogenously determined automation trends in the US and attempt to uncover causal effects on crime rate trends, this paper exploits a SSIV offered by AR (2020). This analysis adds to existing literature studying effects of labour market shocks on crime rates. Through studying a long-run, plausibly exogenous shock, this paper finds no significant immediate impact of industrial automation on the trend in

[5] $Coverage\ ratio_{ct} = (1 - \Sigma_{n=1}^{N} \left\{\frac{Agency\ population_{nt}}{County\ population_{ft}}\right\} * \left[\frac{12-months\ reported_{nt}}{12}\right]\}) * 100$

The degree of 'non-reporting' is increasing in the number of months missing, without imposing as strict an assumption that a full set of 12-monthly observations should be available to compute crime rates.

property and violent crime rates, but a significant, positive effect materialising 7 years later. This increase is consistent with the expectation that automation may adversely affect crime rates due to negative labour market effects of job displacement and increased economic inequality, which existing literature has found to be determinants of crime rates. Although these are likely candidates for mechanisms of the increase, this is merely speculative, since this paper does not formally test causal pathways.

As illustrated by the first stage results and Kleibergen-Paap F-statistic, the instrument satisfies the relevance condition and maintains power with the full set of controls. However, the success of the IV2SLS strategy hinges on the validity of assumptions A1 and A2. Given that the exogeneity of employment shares is unlikely to hold, this paper pursues the shock-identification approach offered by BHJ (2022). This paper found no evidence to reject the randomness of assigned industry-level shocks (A1). Exogeneity is a logical possibility, considering the use of foreign robotics penetration to isolate variation in US robotics adoption. Using countries that are more technologically advanced than the US eliminates the possibility of imitation, thereby reducing the likelihood that the European trend is influenced by US industry-level unobservable factors. A1 has been investigated formally in a simple falsification exercise, which suggested that CZs are not on different trajectories of crime rate shifts, prior to the estimated technological shock in 1993. However, this evidence is restricted to three pre-robotics exposure windows. Ideally, additional 14-year window should be tested, but this is avoided due to limited reliability of crime data prior to 1975. Although it is also not possible to check the validity of A2, future research should seek to indirectly verify the assumption of mean-independent shocks using the tested proposed by BHJ (2022), which tests clustering of residuals from an auxiliary regression of shocks, $\tilde{g}_i$, on industry-level observables, $\bar{X}_i$.

An additional threat to the analysis is the serial correlation of residuals in the main regression equation, induced by similar industry exposure shares across CZs. This concern was first described by Adao et al (2019), who proposed that the correlation can emerge independent of geographic location; hence spatial clustering techniques would not manage this issue. BHJ (2022) shows that the equivalent shock-level regression directly yields exposure-robust SEs, and so future research should report industry-level SEs to avoid potential over rejection of the null hypothesis in the event that sectoral shares are not independent (Adao et al., 2019).

The focus of this paper is to identify aggregate effects of automation on crime rates. However, to better understand both the mechanisms and dynamic nature of these effects, empirical work should investigate shifts in crime rates along the skill distribution. This is important for two reasons. Firstly, a growing body of research, discussed in section 2, finds that low skilled, low wage workers are disproportionately affected by automation, therefore regions with higher shares of low-skill workers may be more susceptible to an immediate increase in the crime rate change. The absence of distributional analysis in this paper could explain the failure to identify significant immediate effects. Secondly, investigating different worker groups might uncover interesting patterns regarding the 'recovery' in criminal activity, as seen in Figures 4 and 5, where the coefficients return to zero around 2002-2016. Recent work by Moretti (2023) finds that following a period of deindustrialisation, areas with a higher share of college-educated workers recover faster in terms of total employment – and in up to 34% of cases, areas adapt and total employment actually increases.

8. Conclusion

The objective of this paper is to apply the SSIV developed by AR (2020) to investigate the effects of automation on employment and wages, to crime rates in local US labour markets. This approach was motivated by endogeneity concerns surrounding US robotics adoption – in particular, the high likelihood of CZ exposure to robotics being determined by local, unobserved economic conditions which also influence a change in crime rates. IV estimates which were substantially larger than OLS in magnitude indicate that the strategy was successful in cleaning the model of some endogeneity. While no immediate effects on crime rates were found during the period of the robotics exposure shock (1993-2007), significant positive effects on property crime rates were observed in subsequent years (2000-2014 and 2001-2015), along with an effect on violent crime rates (2000-2014). This suggests a delayed negative impact of automation on crime rates, possibly reflecting the time it takes for job displacement effects to manifest. However, future research should account for future robotics exposure, to avoid overestimation of this effect. Furthermore, investigating how effects of automation vary across different skill levels in local labor markets will provide valuable insights into policy measures aimed at mitigating potential increases in criminal activity associated with automation-induced job displacement.

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10. Appendix

Table A.1: Means of reporting vs. non-reporting CZs

		Mean	SD
Non reporters	US robotics adoption	1.33	1.23
	Median income ($)	21978.86	4876.13
	Unemployment (%)	5.31	2.85
Reporters	US robotics adoption	1.485	1.41
	Median income	22781.027	4931.168
	Unemployment (%)	5.457	2.911

Table A.2: Robustness check: IV2SLS coefficients (1993-2007)

	Change in (log) property crime rate		Change in (log) violent crime rate	
US robotics exposure	-0.006	-0.012	0.039	0.034
	(0.019)	(0.017)	(0.021)	(0.024)
Kleibergen-Paap rk Wald F statistic	63.73	59.80	63.79	59.88
N	551	551	540	540

** p<.01, * p<.05

Table A.3: Robustness check: IV 2SLS estimates over time

	Change in (log) property crime rates	N	Change in (log) violent crime rates	N
1993-2007	-0.012 (0.017)	551	0.034 (0.024)	540
1994-2008	0.072 (0.051)	534	0.008 (0.078)	525
1995-2009	0.045** (0.013)	541	-0.010 (0.024)	532
1996-2010	-0.006 (0.013)	505	-0.026 (0.022)	500
1997-2011	0.013 (0.001)	516	0.006 (0.019)	507
1998-2012	-0.001 (0.021)	517	-0.038 (0.044)	510
1999-2013	-0.005 (0.015)	527	0.031 (0.025)	520
2000-2014	0.060** (0.023)	563	0.175** (0.065)	554
2001-2015	0.048** (0.016)	556	0.045* 0.020)	547
2002-2016	0.043** (0.016)	561	0.008* (0.019)	553
2003-2017	0.028 (0.011)	552	-0.013 (0.018)	542